

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0484

Placid Lake 138 kV
September 15, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0484. The Interconnection Customer proposes a 268.3 MW solar photovoltaic facility interconnecting at the Placid Lake 138 kV substation in the West Virginia area. The request was received on June 05, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 256 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$112,185,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Dpc Prairie Du Chien to Frito Lay 138 kV	Purdy to Chatom 138 kV	122.8	233	7.3
Placid Lake to Brackenridge 138 kV line	Blackhawk to Placid Lake 138 kV Line	113.1	235	100.0
Brackenridge to Wilsona 138 kV	Blackhawk to Placid Lake Substation 138 kV	112.8	234	100.0
Placid Lake to Blackhawk 138 kV	Brackenridge to Placid Lake 138 kV line	112.7	236	100.0
Wilsona to Roe 138 kV	Placid Lake Substation to Blackhawk 138 kV Line	112.2	234	100.0
Tulloch Tap to Roe 138 kV	Placid Lake to Blackhawk 138 kV Line	111.7	231	100.0
Tulloch Tap to Terrell Hills 138 kV	Placid Lake to Blackhawk 138 kV	111.0	231	100.0
Philpott Substation to Williams Lake 138 kV	Sherburne to Pinetree 345 kV	110.7	200	13.8
Tampa Bay to Roslyn 138 kV	Bath Conty to Sherburne 345 kV	109.8	214	6.0
Philpott to Folsom 138 kV Line	Sherburne to Pinetree 345 kV	109.5	208	13.8
Terrell Hills to N Omaha 138 kV	Blackhawk to PlacidLake 138 kV Line	109.4	224	100.0
Declo to Blackhawk 138 kV Line	Brackenridge to Placid Lake Substation 138 kV	109.2	237	100.0
Fluvanna P.S to Louisa Generation Facility 345 kV	Leavitt to Birchwood Substation 345 kV L	106.6	2,485	8.2
Lilly Del Caribe to Garrion Substation 138 kV	Bath County to Sherburne 345 kV	105.6	251	6.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Bear to Leavitt 345 kV	Bear to Leavitt 345 kV Ckt 2	103.1	1,903	6.2
Mackay to Sherburne 138 kV line	Sherburne to Pinetree 345 kV	102.2	287	8.0
Seminole Electric Cooperative to Van Dyke 138 kV line	Sherburne to Pinetree 345 kV	101.7	281	8.0
Louisa Generation Facility 345/138 kV TX	Fluvanna P.S to Louisa Generation Facility 345 kV	100.6	1,178	6.4
Louisa Generation Facility to N Anna 345 kV line	Louisa Generation Facility to Carrabelle 345 kV	100.2	2,511	9.3
Levitt to Bear 345 kV Circuit 2	Bear Substation to Leavitt 345 kV L	100.2	1,957	6.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Frito Lay Substation to Dpc Prairie Du Chien 138 kV	Rebuild line with higher-capacity conductor	\$3,605,000	48% (\$1,730,000)
Placid Lake to Brackenridge 138 kV line	Rebuild line with higher-capacity conductor	\$9,954,000	100% (\$9,954,000)
Brackenridge Substation to Wilsona 138 kV	Rebuild line with higher-capacity conductor	\$8,880,000	100% (\$8,880,000)
Blackhawk to Placid Lake 138 kV L	Rebuild line with higher-capacity conductor	\$9,796,000	100% (\$9,796,000)
Wilsona to Roe 138 kV	Rebuild line with higher-capacity conductor	\$14,702,000	100% (\$14,702,000)
Tulloch Tap to Roe 138 kV	Rebuild line with higher-capacity conductor	\$10,881,000	100% (\$10,881,000)
Terrell Hills to Tulloch Tap 138 kV line	Rebuild line with higher-capacity conductor	\$11,251,000	100% (\$11,251,000)
Philpott Substation to Williams Lake 138 kV	Rebuild line with higher-capacity conductor	\$9,325,000	75% (\$6,978,000)
Tampa Bay to Roslyn 138 kV	Reconductor line	\$2,724,000	44% (\$1,199,000)
Philpott to Folsom 138 kV	Reconductor line	\$3,550,000	75% (\$2,656,000)
Terrell Hills to N Omaha 138 kV line	Reconductor line	\$2,352,000	100% (\$2,352,000)
Declo to Blackhawk 138 kV Line	Reconductor line	\$16,114,000	100% (\$16,114,000)
Fluvanna P.S to Louisa Generation Facility 345 kV	Reconductor line	\$23,960,000	4% (\$854,000)
Lilly Del Caribe to Garrion Substation 138 kV	Reconductor line	\$5,694,000	100% (\$5,694,000)
Bear to Leavitt 345 kV	Reconductor line	\$86,010,000	4% (\$3,266,000)
Mackay to Sherburne 138 kV line	Reconductor line	\$1,441,000	57% (\$817,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Seminole Electric Cooperative to Van Dyke 138 kV line	Reconductor line	\$3,459,000	57% (\$1,960,000)
Louisa Generation Facility 345-138 kV Transformer Bank	Replace transformer with higher-rated unit	\$10,878,000	2% (\$275,000)
Louisa Generation Facility to N Anna 345 kV line	Reconductor line	\$21,079,000	3% (\$610,000)
Bear to Leavitt 345 kV Cir. 2	Reconductor line	\$58,363,000	4% (\$2,216,000)
	Total allocated to project		\$112,185,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.